

EX.NO : 05

DATE :

CONNECTING TWO LANs USING BRIDGE

AIM :

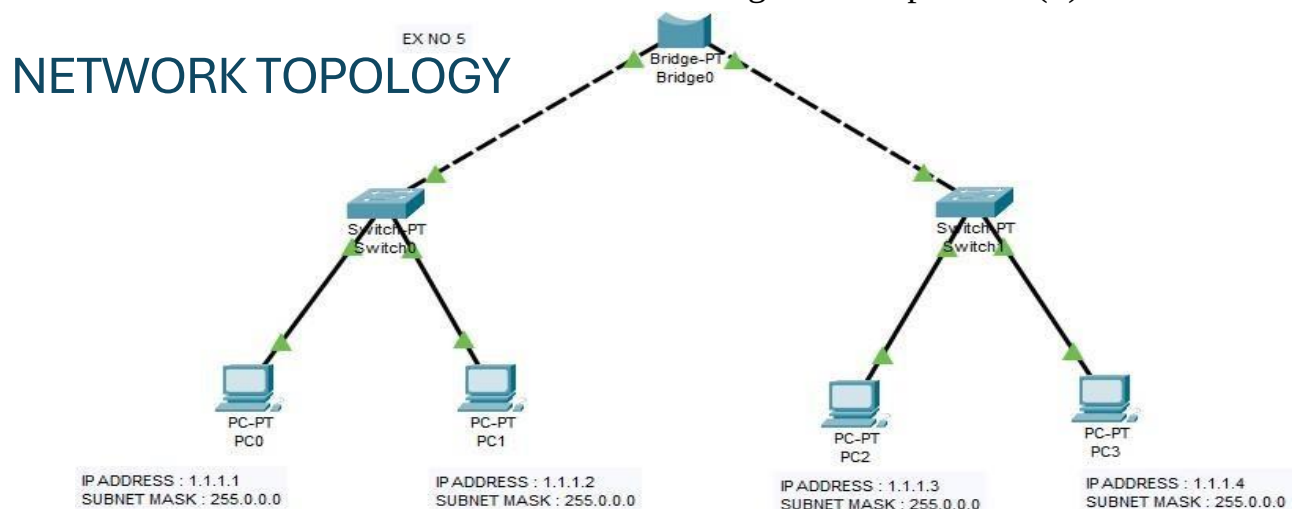
To establish connection between two LANs by using Bridge.

REQUIREMENTS :

- Four windows PC.
- Two Switch (PT).
- Six Straight Line LAN Cables.
- One Bridge (PT).
- Cisco Packet Tracer.

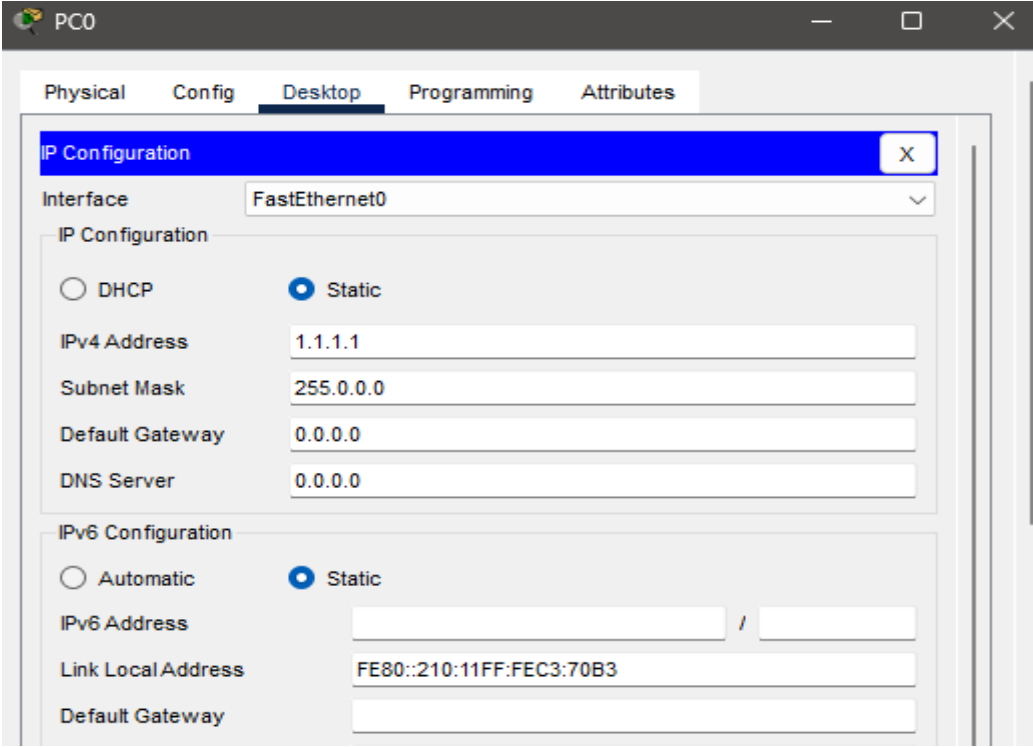
PROCEDURES :

- Open CISCO PACKET TRACER software.
- Draw the Four PC using END Device Icons.
- Draw the Cisco PT Port Switch Using Switch icon lists.
- Draw the Cisco PT Bridge.
- Make the Connections using Copper-Straight-Through Ethernet Cables.
- Enter the IP Address To Each Machine.
- Check the Network Connections using Add Simple PDU(P).



HOST PC IP ADDRESS :

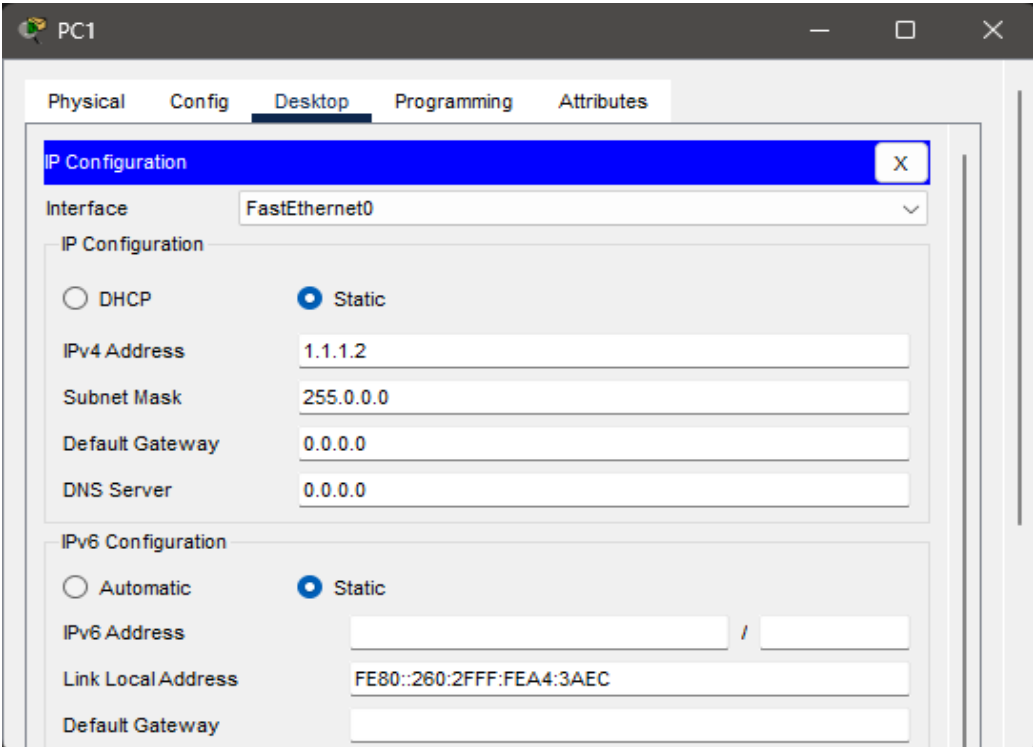
PC0 IP CONFIGURATION



The image shows the 'PC0' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', 'Static' is selected. The 'IPv4 Address' is '1.1.1.1', 'Subnet Mask' is '255.0.0.0', 'Default Gateway' is '0.0.0.0', and 'DNS Server' is '0.0.0.0'. Under 'IPv6 Configuration', 'Static' is selected. The 'IPv6 Address' field is empty, 'Link Local Address' is 'FE80::210:11FF:FEC3:70B3', and 'Default Gateway' is empty.

Field	Value
Interface	FastEthernet0
IP Configuration	Static
IPv4 Address	1.1.1.1
Subnet Mask	255.0.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	Static
IPv6 Address	
Link Local Address	FE80::210:11FF:FEC3:70B3
Default Gateway	

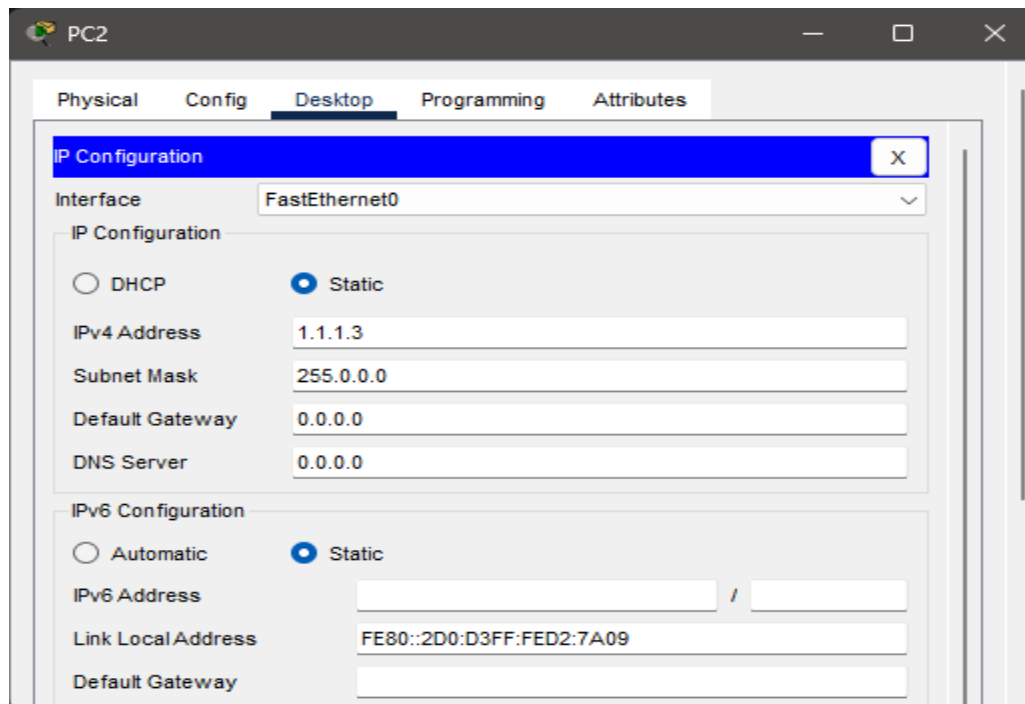
PC1 IP CONFIGURATION



The image shows the 'PC1' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', 'Static' is selected. The 'IPv4 Address' is '1.1.1.2', 'Subnet Mask' is '255.0.0.0', 'Default Gateway' is '0.0.0.0', and 'DNS Server' is '0.0.0.0'. Under 'IPv6 Configuration', 'Static' is selected. The 'IPv6 Address' field is empty, 'Link Local Address' is 'FE80::260:2FFF:FEA4:3AEC', and 'Default Gateway' is empty.

Field	Value
Interface	FastEthernet0
IP Configuration	Static
IPv4 Address	1.1.1.2
Subnet Mask	255.0.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	Static
IPv6 Address	
Link Local Address	FE80::260:2FFF:FEA4:3AEC
Default Gateway	

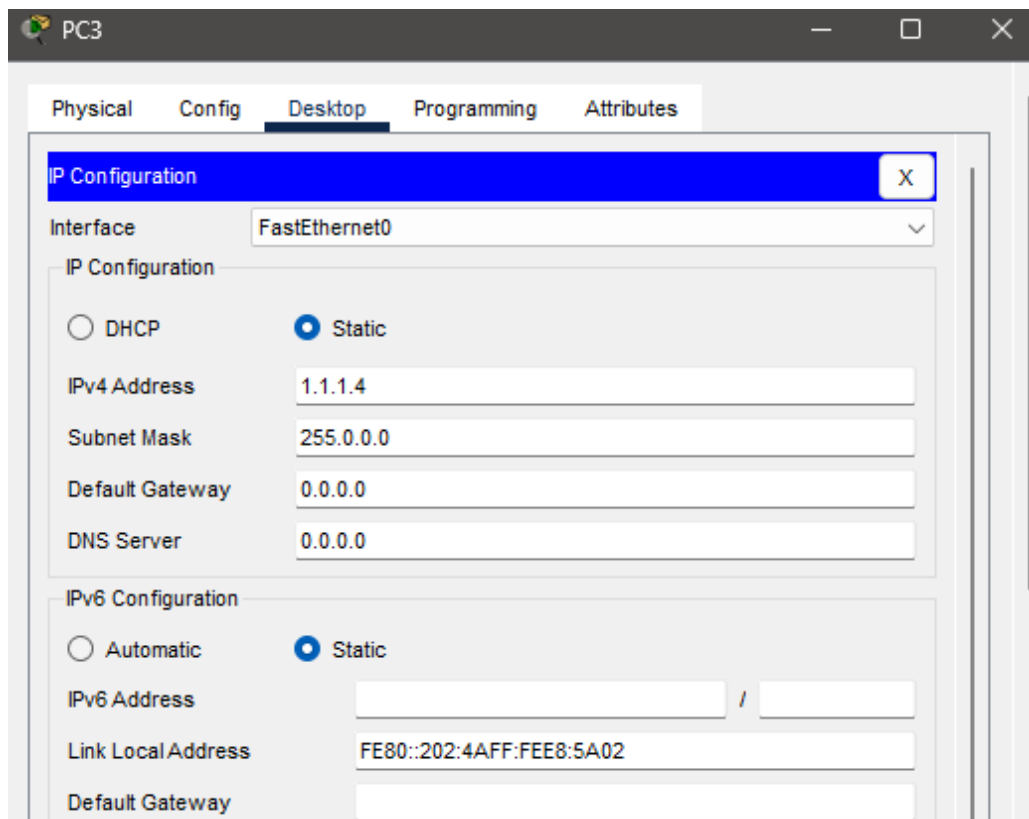
PC2 IP CONFIGURATION



The screenshot shows the 'PC2' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' dropdown is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The IPv4 fields are filled with: IPv4 Address: 1.1.1.3, Subnet Mask: 255.0.0.0, Default Gateway: 0.0.0.0, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is also selected. The IPv6 fields are: IPv6 Address (empty), Link Local Address: FE80::2D0:D3FF:FED2:7A09, and Default Gateway (empty).

Section	Option	Field	Value
IP Configuration	☒ Static	IPv4 Address	1.1.1.3
		Subnet Mask	255.0.0.0
		Default Gateway	0.0.0.0
		DNS Server	0.0.0.0
IPv6 Configuration	☒ Static	IPv6 Address	
		Link Local Address	FE80::2D0:D3FF:FED2:7A09
		Default Gateway	

PC3 IP CONFIGURATION



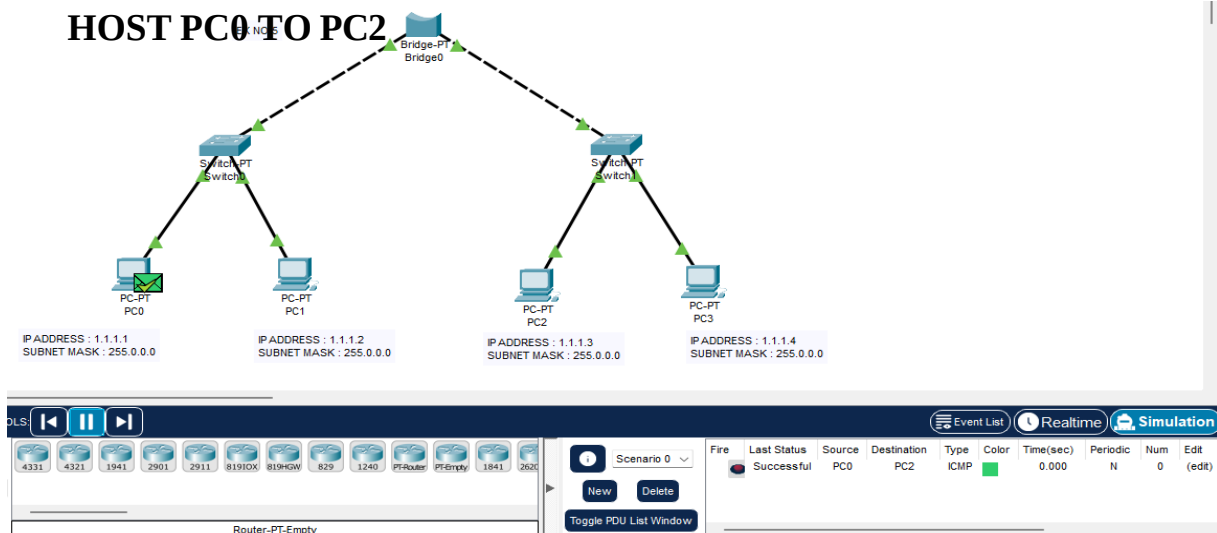
The screenshot shows the 'PC3' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' dropdown is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The IPv4 fields are filled with: IPv4 Address: 1.1.1.4, Subnet Mask: 255.0.0.0, Default Gateway: 0.0.0.0, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is also selected. The IPv6 fields are: IPv6 Address (empty), Link Local Address: FE80::202:4AFF:FEE8:5A02, and Default Gateway (empty).

Section	Option	Field	Value
IP Configuration	☒ Static	IPv4 Address	1.1.1.4
		Subnet Mask	255.0.0.0
		Default Gateway	0.0.0.0
		DNS Server	0.0.0.0
IPv6 Configuration	☒ Static	IPv6 Address	
		Link Local Address	FE80::202:4AFF:FEE8:5A02
		Default Gateway	

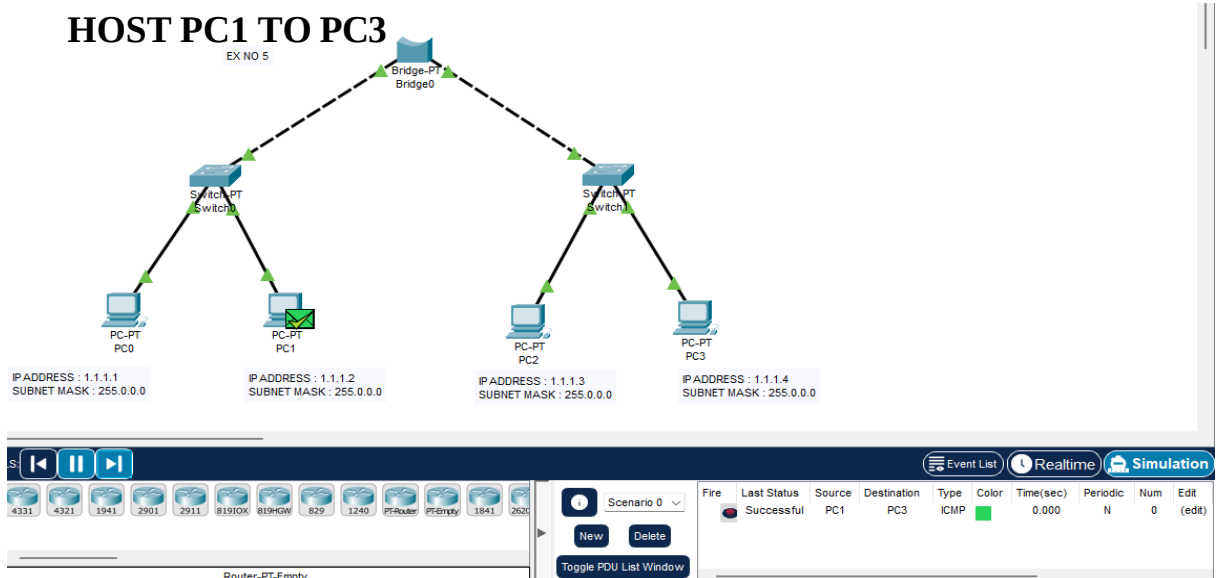
VERIFY LAN NETWORK CONNECTIVITY :

Using Add Simple PDU(p), Click the mail icon and then drop one mail to one of the PC in first lan and another mail to PC in another lan. If the resultant window shows the successful delivery, then network connectivity is successful.

HOST PC0 TO PC2



HOST PC1 TO PC3



RESULT :

Thus, two LANs are connected using Bridges and the communication between LANs is checked successfully.

